

MLOps Lab 1 – EC-1 Lab Assignment

Experiment Tracking and Model Registry with MLflow

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1. Screenshot: Experiment Table / Run Comparison

MLflow experiment evidence showing the CartVista churn runs and their tracked parameters and metrics.

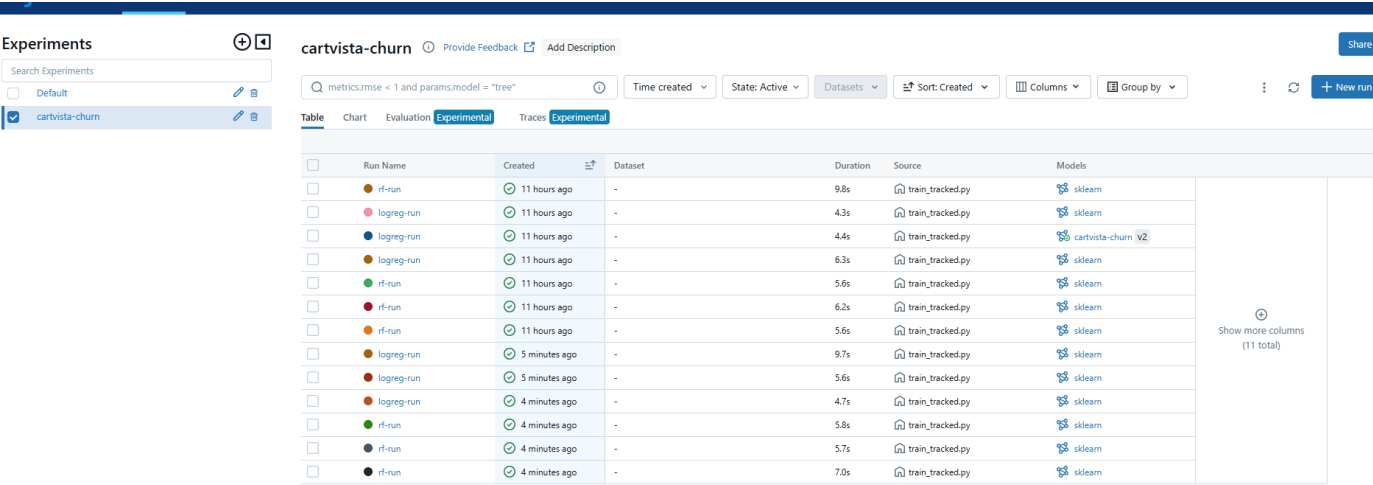


Figure 1. MLflow experiment table containing tracked runs.

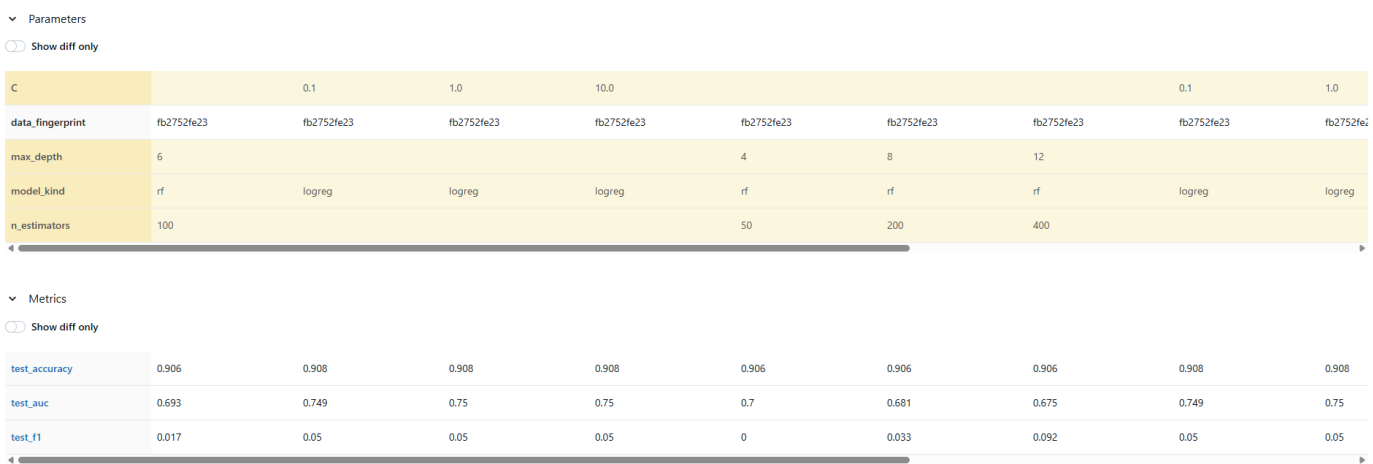


Figure 2. MLflow parameter and metric comparison, including test_auc.

2. Screenshot: Models Page – Versions and @champion

The registered model page shows cartvista-churn with model versions and the @champion alias.

Registered Models >

cartvista-churn

Created Time: 2026-09-12 22:23:46Last Modified: 2026-09-13 10:10:15

> Description

Edit

> Tags

▼ Versions

Compare

New model registry UI

Version	Registered at	Created by	Tags	Aliases	Description
☒ Version 3	2026-09-13 10:10:15		Add	@ champion ✎	
☐ Version 2	2026-09-12 22:24:10		Add	Add	
☒ Version 1	2026-09-12 22:23:46		Add	Add	

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Figure 3. cartvista-churn model versions with @champion alias.

Registered Models

Create Model

Filter registered models by name or tags

Name	Latest version	Aliased versions	Created by	Last modified	Tags
cartvista-churn	Version 3	@ champion : Version 3		2026-09-13 10:10:15	—

Figure 4. Registered Models page showing @champion → Version 3.

3. Terminal Output: consume_model.py

The consumer program loads the registered champion model and produces churn predictions for five customers.

```
Registered cartvista-churn v1 and set alias @champion
(.venv) (base) s2025ae05230@ip-10-3-16-225:~/workspace/cartvista-mlops$ python consume_model.py
/home/s2025ae05230/workspace/cartvista-mlops/.venv/lib/python3.12/site-packages/mlflow/utils/requirements_utils.py:20: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
  import pkg_resources # noqa: TID251
customer_000 -> churn prediction: 0
customer_001 -> churn prediction: 0
customer_002 -> churn prediction: 0
customer_003 -> churn prediction: 0
customer_004 -> churn prediction: 0
(.venv) (base) s2025ae05230@ip-10-3-16-225:~/workspace/cartvista-mlops$
```

Figure 5. consume_model.py terminal output.

4. Written Answers

Step 2 Checkpoint – Why is accuracy high while AUC is lower?

The churn dataset is imbalanced, meaning most customers do not churn. Therefore, a model can obtain high accuracy by predicting the majority class for most customers, even if it is not very effective at distinguishing churners from non-churners. AUC measures the model's ability to rank and discriminate between the two classes, so it gives a more useful view of model discrimination for this imbalanced problem. **Observed result:** AUC = 0.7001 and accuracy = 0.9064.

E1 – Precision and Recall

Precision is particularly important for CartVista because retention offers cost money to send. Precision tells the team what proportion of customers predicted as churners are actually churners, so higher precision reduces wasted offers. Recall is also important because failing to identify a true churner can result in losing that customer. Because the offers have a direct cost, the retention team should give strong emphasis to precision while using recall as a complementary measure and selecting a suitable operating threshold.

E3 – Why are tags the right home for author and purpose?

Tags are appropriate for author and purpose because they are descriptive metadata about a run. Parameters such as C, n_estimators and max_depth describe the model configuration used for training, whereas author and purpose provide context for organizing, filtering and understanding experiments. Therefore, author and purpose belong naturally in MLflow tags rather than model parameters.

E4 – What does the data fingerprint prevent?

The data_fingerprint identifies the data snapshot used for an experiment. If the dataset changes, such as when one row is deleted, the fingerprint changes. This prevents a Priya-vs-Arjun dispute in which both people claim that their models were evaluated on the same data even though their experiments actually used different data snapshots.

5. Reflection

If Priya and Arjun each run MLflow on their own laptops, they will have separate tracking stores and model registries. There will be no single shared source of truth for experiments, model versions or the @champion alias, making it difficult to compare runs, reproduce another person's experiment, or know which model is currently intended for use. A shared MLflow tracking and registry backend is therefore needed for team-level collaboration and reproducibility.